INJECTIVES IN QCOH OF A NOETHERIAN SPACE ARE SUMMANDS OF COLIMITS OF ARTINIAN PUSHFORWARDS

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Editorial note. Counted span is the author's own proof environment, source lines 2880-2895, immediately following the statement at 2867-2878.

Editorial note. The splitting uses only injectivity of J (mono from an injective object splits); the source's phrasing 'injective morphism' is read as monomorphism.

Allowed standard prerequisites (editorial).

- Affine etale cover of a quasi-compact space. Properties of Spaces, Lemma spaces-properties-lemma-quasi-compact-affine-cover, cited at proof lines 2882-2884. Supplies the surjective etale j : U -> X from a Noetherian affine scheme U.
- The affine scheme case. Lemma lemma-affine-injective-colimit-direct-sum-pushforwards-artin of this chapter, cited at proof lines 2894-2895. Terminal input: the corresponding description of injectives on the Noetherian affine scheme U.

Source reference. Original lemma 09TI: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 2831–2831. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03H6: focused-seeds/geometry/S9-perf-injective-colimit-direct-sum-pushforwards-artin/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be an affine scheme. Let X be a Noetherian algebraic space over S. Every injective object of $QCoh(\mathcal{O}_X)$ is a direct summand of a filtered colimit $\operatorname{colim}_i \mathcal{F}_i$ of quasi-coherent sheaves of the form*

$$\mathcal{F}_i = (Z_i \to X)_* \mathcal{G}_i$$

where Z_i is the spectrum of an Artinian ring and $\mathcal{G}_i$ is a coherent module on Z_i.

Proof. Choose an affine scheme U and a surjective étale morphism $j : U \to X$ (Properties of Spaces, Lemma 03H6). Then U is a Noetherian affine scheme. Choose an injective object $\mathcal{J}'$ of $QCoh(\mathcal{O}_U)$ such that there exists an injection $\mathcal{J}|_U \to \mathcal{J}'$. Then

$$\mathcal{J} \to j_* \mathcal{J}'$$

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is an injective morphism in $QCoh(\mathcal{O}_X)$, hence identifies $\mathcal{J}$ as a direct summand of $j_*\mathcal{J}'$. Thus the result follows from the corresponding result for $\mathcal{J}'$ proved in Lemma 09TI. $\square$